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Prabhu Kiran Vandranki

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EDUCATION

Master Of Science — ***Clarkson University*** **Jan 2024 – May 2025**
Major *- Applied Data Science* ***Overall GPA: 3.95/4.00***
Bachelor of Science & Technology — ***Parul University*** **June 2019 – May 2024**
Major *- Computer Science & Engineering* ***Overall GPA: 8.08/10.0***

ACADEMIC & SELF PROJECTS

Global Organ Transplant Procurement Network Management Database System Mar 2024 – May 2024

- Developed a centralized database platform to manage and track organ donors, recipients, and medical professionals, streamlining end-to-end operations for organ transplants.
- Engineered complex relational schemas and optimized queries to enable rapid organ matching and precise tracking of donation and transplant activities.
- Implemented automated donor updates, event logging, and report generation using SQL triggers for real-time system reliability and accuracy, and also used procedures.
- Presented project outcomes and technical documentation, demonstrating effective problem-solving and stakeholder communication in a high-impact healthcare technology setting.

Text Mining & AI Agents Driven Optimization Of Ceria Slurry for CMP | Research Associate **Sept 2024 – Present**

- Conducting research on **Ceria synthesis**, leveraging AI and ML to extract synthesis parameters from 300+ scientific research papers. Developed a program to extract key experimental parameters from XML and PDF sources. Created a structured JSON dataset for accurate model training and fine-tuned custom LLMs (OpenAI, Mistral, Gemini) for automated knowledge extraction.
- Developed a SciBERT model to extract and analyze key experimental parameters, enhancing the accuracy of knowledge extraction in Ceria synthesis. Applied deep reinforcement learning to optimize **CMP slurry formulations**, significantly improving selectivity, defect reduction, and removal rates. drawn and visualized insights & trends, Overall Indepth Data Analysis.

Atomic-Level Insights & Machine Learning for Corrosion Inhibitors in Copper CMP | Sponsored by **IBM** **Jan 2025 – July 2025**

- Designed and implemented data-processing pipeline, incorporating canonical SMILES generation, DFT-derived quantum descriptors, virtual sample augmentation, and RDKit structural molecular fingerprint extraction for feature engineering.
- Developed and evaluated 6 machine learning models, achieving highest validation R^2 of 0.80 with XGBoost. Conducted metric-based model selection and SHAP analysis to interpret key features driving corrosion inhibition efficiency.

Student Chatbot | Team Leader - NATURAL LANGUAGE PROCESSING & MULTINOMIAL NAIVE BAYES ALGO **Oct 2022 – Apr 2023**

- Developed this chatbot to serve as a personalized tutor, identifying areas of weakness in students' understanding and guiding them toward mastery. This bot identifies the student's interests, recommends various books, websites, and videos, conducts exams on the topic, provides analysis, and delivers feedback. This chatbot comprises seven modules, making it a standout bot.

Research Group Operations and Event Management (SEO Research Group, CAMP, Clarkson University) May 2025 – Present
Led critical administrative and logistical functions for one of the largest research groups at CAMP, including:

- Coordinated travel arrangements (flight, hotel, ground transport) for research group members attending conferences and industry seminars.
- Managed advisor and lab tools calendars, scheduling key meetings, events, and resources for seamless team operations.
- Organized room setups, devices, and catering services for professional and academic events, improving stakeholder and attendee experience.
- Facilitated communication between research group members, external industry partners, and service providers to ensure successful collaborative events.
- Maintained budgetary oversight for lab expenses, purchasing, and vendor negotiation.

Stakeholder Outreach and Community Engagement | Research Associate May 2025 – Present
Played an active role in expanding group visibility and building relationships with external partners:

- Participated in business association networking, conference panels, and trade show logistics for the department.
- Presented technical solutions and research outcomes to academic and non-academic audiences.
- Contributed to successful grant proposal writing and project negotiations with industry stakeholders.

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SKILLS

Languages & Tools	Python, Rstudio, Tableau, PowerBI, C, C++, ReactJS, JavaScript, SQL, Kotlin, HTML, CSS, TypeScript
Technologies	Transformers, HuggingFace, LangFlow, Cryptography, OpenCV, Figma, Scikit-Learn, AWS, Git & Github
Quantitative Research	CMP, Text Mining, Statistical Analysis, NLP, RAG, LLMs, BERT, Fine-Tuning, ML, DL, LangChain, ETL
Business & Productivity	Microsoft Office, Google Workspace (G-Suite), Calendar Scheduling (Google Calendar, MS Outlook)
Professional Skills	Project Coordination, Event Planning & Logistics, Account Operations, Technical Solution Presentation, Report Preparation, Community Engagement, Intrastate Travel
Technical & Soft Skills	Windows, Linux, Excel for Data Analysis, Problem-Solving, Programming, Communication, Teamwork, Quick Adaptation to New Tools

TECHINICAL EXPERIENCE

Clarkson University **May 2025 — Present**
*Research Associate — **Seo Research Group, CAMP***

- Designed and deployed an AI-Agent–driven literature mining pipeline that autonomously scraped, parsed, and structured data from 300+ peer-reviewed CMP papers into a machine-readable JSON knowledge base.
- Engineered automated NLP workflows to extract ceria synthesis conditions, slurry chemistry, particle processing parameters, and CMP performance metrics with high accuracy and minimal human intervention.
- Performed large-scale statistical and correlation analysis on thousands of text-mined data points to uncover robust relationships between slurry variables, process settings, and planarization outcomes.
- Established a new data-centric framework for CMP by unifying scattered literature into an actionable database, enabling evidence-based slurry optimization, previously hidden pattern discovery, and improved wafer-level performance predictions.

Clarkson University **Sept 2024 — April 2025**
*Graduate Research Assistant — **Seo Research Group, CAMP***

- Developed a multi-stage research pipeline integrating electrochemical experiments, DFT quantum-chemical calculations, and ML modeling to study structure–property relationships of 50+ organic Cu corrosion inhibitors.
- Designed end-to-end data engineering workflows, including SMILES standardization, RDKit fingerprint extraction, DFT descriptor computation, and virtual sample augmentation for model-ready datasets.
- Built, trained, and evaluated six ML models for predicting inhibitor performance, achieving a top validation accuracy of $R^2 = 0.80$ using XGBoost, and performed SHAP analysis to interpret key molecular features.
- Conducted AI-driven research on ceria nanoparticle synthesis by extracting experimental parameters from 300+ XML/PDF papers, generating structured JSON datasets, and fine-tuning domain-specific LLMs (OpenAI, Mistral, Gemini) and SciBERT for automated scientific knowledge extraction and trend discovery.

TATHASTU / **June 2022 — July 2022**
Full Stack, M.E.R.N SCHOLAR INTERN

- During my internship, I utilized the M.E.R.N stack for frontend, backend, and database. Crafting engaging UIs with React.js, robust backend services with Node.js, and ensuring data integrity with MongoDB and Mongoose.js. Additionally, I integrated e-commerce functionalities, conducted testing, and orchestrated cloud deployment with proactive monitoring.
- I contributed to the creation of an innovative E-commerce platform specializing in clothing and fashion accessories.

LETS GROW MORE / **August 2022 — September 2022**
ANDROID DEVELOPMENT

- During my internship, I developed a Budget Tracker app, honing my proficiency in Java and Kotlin. This project allowed me to implement features for income, expenses, and savings tracking, empowering users to take control of their financial health. Through hands-on Android development work, enhancing my problem-solving skills and overall app development expertise.

AWARDS & ACHIEVEMENTS

Attended multiple conferences and Presented AI&ML-driven **CMP semiconductor** research projects includes at **Lewis University's CMPUG Conference** in **Illinois**, the **CAMP Technical Meeting Conference** in **Geneva**, and the **27th CAMP CMP International Symposium** in **Saratoga**, highlighting abstracts, methodologies, findings, and **industry relevance**.

wrote **2** research papers:one on a **student chatbot** Undergrad capstone project and another on a **talented assistance** self project.

Solved 50+ problems on **Leetcode** in Python & c++, and Worked with **Chegg** as a math expert for **Problem Sloving**.

Certified by **LinkedIn** for completing **JavaScript** course. Passed **LinkedIn Skill Assessment** in Python, Excel, and MSOffice.